

Article

Mobility-Based Multi-Hop Content Precaching Scheme in Content-Centric Vehicular Networks

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Abstract: Due to the rapid development of smart vehicles, such as self-driving cars, the demand for mobile data traffic by vehicle users has increased so much that base stations cannot handle it, causing delays in content provision. The burden on the base station can be alleviated through roadside units (RSUs) to distribute the demand. However, outage zones, which fall outside the communication range of RSUs, still exist due to their high deployment cost. Existing schemes for covering outage zones have only considered single-hop precaching vehicles to provide precached content, which is insufficient to reduce outage zones effectively. Therefore, we propose a scheme to reduce outage zones by maximizing the amount of precached content using multi-hop precaching vehicles. The proposed scheme optimally selects precaching vehicles through a numerical model that calculates the amount of precached content. It enhances the process of multi-hop precaching by comparing the connection time of vehicles with the dark area time in the outage zone. To prevent excessive overheads due to frequent precaching vehicle handovers, the proposed scheme limits the selection to vehicles with a longer communication time, based on a precaching restriction indicator in the multi-hop precaching vehicle selection process. The simulation results show that our scheme outperforms representative schemes based on single-hop precaching.

Keywords: content-centric vehicular networks;  content precaching; multi-hop precaching



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1. Introduction

With the rapid development of wireless communication and vehicular manufacturing technologies, smart vehicles, such as self-driving cars, have become more realistic than ever, thanks to the efforts of many car companies and research institutions [1]. Since passengers, including drivers, no longer need to focus on driving while traveling to their destinations with the help of smart vehicles, they have more free time to enjoy various applications, such as entertainment, gaming, and multimedia advertising, through vehicular networks [2]. As a result, the demand for content from these applications continues to grow. Furthermore, as the quality of content improves, its size also increases. The rising demands and larger content sizes are causing mobile data traffic to increase dramatically in vehicular networks. This surge significantly burdens the base stations (BSs) that cover wide areas on roads through cellular communications. Due to the capacity limitations of BSs, the quality of service (QoS) for vehicle users may drop significantly, leading to long delays and buffering, similar to the communication issues experienced in crowded places [3].

Many researchers have studied Content-Centric Vehicular Networks (CCVNs) to address user requests in a distributed manner by using roadside units (RSUs), which have small coverage areas and cache-forwarded and -provided content in their storage [4–9]. RSUs have sufficient wireless communication resources because they cover small areas using communication technologies such as WAVE or WiFi. Additionally, RSUs can reduce

access delay and traffic consumption when retrieving requested content, as some content is already cached in their storage. However, when content that is not cached in an RSU's storage is requested, the RSU must access the content server to obtain the content. This access adds an additional delay and generates backhaul link traffic. Furthermore, due to the high deployment cost and limited communication range of RSUs, they cannot cover the entire area on roads in CCVNs. As a result, an outage zone occurs between two neighboring RSUs, where no RSU provides coverage. This outage zone must be covered by cellular communication through base stations (BSs), which incurs high communication costs for vehicle users. Moreover, this issue leads to degraded QoS for content delivery to vehicle users in CCVNs.

To address these issues, many researchers have studied precaching, which involves proactively caching requested content in vehicles that a requester vehicle can communicate with within the outage zone [10–16]. This approach allows the requester vehicle to receive the requested content from precaching vehicles through direct communication, without relying on base stations (BSs) in the outage zone. In this paper, we define vehicles that precache the requested content as precaching vehicles for requester vehicles. However, existing schemes for selecting precaching vehicles in the outage zone do not achieve optimized delay performance because they only use single-hop precaching vehicles without considering multi-hop connections. If a single-hop precaching vehicle has a long connection time with the requester vehicle but a short precaching time from an RSU, it cannot effectively leverage its long connection time for precaching due to the short precaching time. As a result, it does not significantly contribute to reducing the outage zone. Furthermore, due to the low vehicle density, selecting a small number of precaching vehicles within the single-hop communication range of the requester vehicle cannot sufficiently reduce the outage zone. Fortunately, the remaining connection time, aside from the precaching time, can be utilized to connect to other vehicles in the multi-hop communication range through the single-hop precaching vehicle. With multi-hop connections, these vehicles can also be selected as precaching vehicles for content precaching, further reducing the outage zone for the requester vehicle.

Therefore, we propose a multi-hop precaching scheme that effectively reduces outage zones by utilizing vehicles in the multi-hop communication range as precaching vehicles in CCVNs. First, the proposed scheme uses a numerical model to determine the need to precache the requested content from the requester vehicle in the outage zone. If precaching is necessary, the RSU calculates the amount of requested content that needs to be precached by considering the mobility of each candidate vehicle within its communication range. The RSU then selects multiple vehicles to ensure the entire amount of requested content can be precached. Once enough vehicles are selected to ensure that all of the requested content can be delivered in the outage zone, the RSU precaches the content to these vehicles. Next, to utilize multiple precaching vehicles, the proposed scheme extends the communication range to multi-hop if the RSU cannot find enough precaching vehicles within the single-hop communication range of the requester vehicle. Since the RSU can obtain the mobility information of each vehicle, it calculates the connectivity between the requester vehicle and those located within the multi-hop communication range, using them as precaching vehicles for the requested content. However, in multi-hop communication, connectivity significantly decreases as the hop count increases, and overhead rises. Therefore, the proposed scheme limits the maximum number of hops to three [17]. To minimize the delay overhead caused by frequent handovers between the precaching vehicle and the requester vehicle due to the multi-hop communication characteristics, we define a precaching limit metric called *minimum selection time* in the multi-hop precaching vehicle selection process. The proposed scheme avoids inefficient handovers by not selecting a candidate vehicle as a precaching vehicle if its connection time is shorter than the *minimum selection time*. Simulations are conducted to compare the performance of the proposed scheme with two approaches: Adaptive Carry–Store–Forward (ACSF) [10], and Set Ranking-based Precaching (SRP) [16]. The simulation results show that the proposed scheme performs better than the two existing

schemes in terms of connectionless time and dark area time during outage zones, which are important metrics for content precaching.

Our contributions are as follows:

- The proposed scheme introduces a quantitative model to determine the necessity of content precaching for requester vehicles in the outage zone, using the mobility information of vehicles, such as vehicle trajectories, speeds, and positional variability within the RSU's communication range. Based on this model, the RSU optimizes the selection of multiple precaching vehicles, ensuring that the requested content is efficiently distributed across these precaching vehicles to minimize communication failure and guarantee reliable content delivery within the outage zone.
- The scheme extends content delivery capabilities by incorporating multi-hop communication, allowing the RSU to dynamically calculate the connectivity of vehicles within a multi-hop communication range using real-time mobility information. This enables the RSU to select vehicles beyond the single-hop range as precaching vehicles, significantly expanding the communication range while maintaining reliable connectivity. A connectivity evaluation algorithm assesses the stability of connections up to three hops to ensure efficient content delivery without excessive overheads.
- The proposed scheme introduces the **minimum selection time** metric to reduce the delay caused by frequent handovers in multi-hop communication. This metric ensures that only vehicles with stable, long-duration connections are selected as precaching vehicles, reducing the number of unnecessary handovers. This approach decreases handover occurrences, delays overheads, and ensures smooth content transmission.
- Simulations are conducted to compare the proposed scheme with two existing schemes: Adaptive Carry–Store–Forward (ACSF) and Set Ranking-based Precaching (SRP) for performance evaluation. The results of the simulation, conducted in various network environments, demonstrate that the proposed scheme outperforms both ACSF and SRP in terms of the connectionless time and dark area time during outage zones.

The remainder of this paper is organized as follows. Section 2 reviews related works. Section 3 describes the network model and provides an overview of the proposed scheme. The details of our multi-hop precaching scheme are presented in Section 4. The simulation results are discussed in Section 5, demonstrating the performance of the proposed scheme. Finally, Section 6 concludes the paper.

2. Related Works

In Vehicular Ad Hoc Networks (VANETs), vehicular communication is closely related to connectivity, latency, and the quality of experience (QoE) for content delivery to vehicle users. However, VANETs have inherent weaknesses, such as packet loss and delays, due to the increased demand for large amounts of content, as they rely on the IP-centric networking approach. To address these challenges, Content-Centric Vehicular Networks (CCVNs) have emerged, enabling VANETs to adopt the Content-Centric Network (CCN) approach [18,19]. In CCNs, every node stores and serves a portion of the content by utilizing caching capabilities [5–9,12,20–23]. In CCVNs, RSUs and vehicles act as nodes to store and deliver content to requester vehicles. RSUs access servers via backhaul links to reduce network traffic and enable fast content delivery. However, the high installation cost and limited communication range of RSUs can result in outage zones between neighboring RSUs [24]. In these outage zones, requester vehicles cannot download content from any RSU, as they fall outside the communication coverage of any RSU. To address this issue, research has focused on enabling vehicles on the road to deliver content through precaching [9,25]. With the help of vehicle precaching, CCVNs can compensate for the communication coverage gaps affecting requester vehicles in outage zones, thereby providing efficient content delivery.

Thus, we review existing research on content delivery using precaching vehicles for outage zones in CCVNs. Some studies [10,14,16,26] have explored using vehicles moving in the same direction as requester vehicles as precaching vehicles in outage zones. First,

Wu et al. [10] proposed an Adaptive Carry–Store–Forward (ACSF) scheme that selects a passing vehicle to precache content for a requester vehicle as it passes through the RSU location. To maximize communication time within the outage zone, the requester vehicle adjusts its speed relative to the precaching vehicle using the internal point method. Despite its advantages, ACSF has limitations, as it uses only a single precaching vehicle, which restricts the amount of content that can be delivered within the outage zone. Additionally, it can be impractical for the requester vehicle to adjust its speed to match that of the precaching vehicle. Bang et al. [14] proposed a similar scheme that selects a single precaching vehicle for a requester vehicle in an outage zone. In this approach, the precaching vehicle is chosen based on the downloading and relaying times of content, using correlated mobility information between the requester and candidate vehicles. However, like ACSF, this scheme suffers from the same limitation of relying on a single precaching vehicle.

To overcome the limitations of using a single precaching vehicle, several studies [16,26,27] have explored the use of multiple precaching vehicles in outage zones. First, Bouk et al. [27] proposed a bivious selection scheme that selects two precaching vehicles (i.e., a backward precaching vehicle and a forward precaching vehicle) for a requester vehicle in an outage zone. The backward precaching vehicle is chosen from the backward candidate region by the current RSU after the requester vehicle leaves its coverage. In contrast, the forward precaching vehicle is selected from the forward candidate region by the next RSU before the requester vehicle enters its coverage. Ahmed et al. [26] proposed a scheme for selecting more than two precaching vehicles in outage zones. This scheme enables the RSU to select multiple precaching vehicles by evaluating various properties of vehicles within the one-hop communication range of the requester vehicle. The RSU ranks each potential precaching vehicle based on its connection time and available caching capacity, distributing the amount of content to be precached accordingly. However, this scheme has a drawback: when the requester vehicle requests content, the top-ranked precaching vehicles use their entire caching capacity, leaving them unable to directly download content from RSUs for themselves. To prevent the issue of overloading top-ranked vehicles, Nam et al. [16] proposed the Set Ranking-based Precaching (SRP) scheme, which selects a group of multiple precaching vehicles for outage zones. This scheme calculates all possible precaching vehicle sets and ranks them based on their available caching capacities. The final content precaching group is selected by considering both the set rankings and vehicle communication overheads. However, since these schemes select vehicles moving in the same direction as the requester vehicle, they encounter a problem when no neighboring vehicle moving in the same direction is available within the communication range of the requester vehicle.

To address the issue of not selecting a precaching vehicle moving in the same direction as the requester vehicle, several studies have proposed schemes [11,28] that utilize vehicles moving in the opposite direction as precaching vehicles. These schemes take advantage of the fact that vehicles coming from the opposite direction will inevitably meet the requester vehicle. Chen et al. [28] proposed a scheme that uses one vehicle moving from the opposite direction as a precaching vehicle. In this scheme, when a requester vehicle requests content from the current RSU, the RSU asks the next RSU (where the requester vehicle will arrive next) to select a content precaching vehicle. The next RSU then selects one vehicle within its communication range that is moving toward the current RSU, and downloads the content for the requester vehicle to this precaching vehicle. Because the precaching vehicle will inevitably encounter the requester vehicle in the outage zone, it can deliver the precached content. To utilize multiple precaching vehicles moving in the opposite direction, Guo et al. [11] proposed a scheme in which the next RSU forms a linear cluster of multiple vehicles within its communication range. These vehicles collectively download the content for the requester vehicle from the RSU as precaching vehicles. The average data volume that each precaching vehicle in the linear cluster can download depends not only on its sojourn time T within the coverage of the next RSU but also on the number of precaching vehicles concurrently present within that coverage. The requester vehicle continues to download content from each precaching vehicle in the linear cluster while

in the outage zone. However, despite the advantage of inevitably meeting the requester vehicle, precaching vehicles moving in the opposite direction face the limitation of low content delivery capacity due to their short communication time with the requester vehicle.

To enhance the efficiency of content delivery in the outage zone, research has focused on using precaching vehicles in both the same and opposite directions. Wang et al. [12] proposed the Cooperative Store–Carry–Forward (CSCF) scheme, which selects one precaching vehicle in the same direction and one in the opposite direction through a two-stage process to increase the amount of content delivered to a requester vehicle in outage zones. This selection involves cooperation between two consecutive neighboring RSUs connected via backhaul links. The first RRU selects the initial precaching vehicle moving in the same direction as the requester vehicle and sends relevant information, such as the communication time between the initial precaching vehicle and the requester vehicle, to the next RRU. The next RRU then selects a second precaching vehicle moving in the opposite direction based on the information received. However, since this scheme relies on only two precaching vehicles, it may not provide sufficient content to requester vehicles in outage zones. Additionally, it inherits the limitations of precaching schemes that utilize vehicles in both the same and opposite directions.

However, the above-examined precaching schemes still have limitations in reducing connectionless time and dark area time in the outage zone. Most notably, these schemes rely on single-hop communication between a requester vehicle and precaching vehicles, both in the same and opposite directions. As a result, if there are only a few vehicles available as precaching vehicles within the single-hop communication range of a requester vehicle, only a limited number of precaching vehicles can be selected in the outage zone, which is insufficient to reduce dark area time and connectionless time effectively. Additionally, even if a single-hop precaching vehicle has a long connection time with the requester vehicle, much of its connection time—aside from the time spent delivering its precached content—cannot be utilized for further content delivery to the requester vehicle. Therefore, this paper aims to increase the number of precaching vehicles available to the requester vehicle by utilizing multi-hop communication between vehicles, thereby reducing dark area time and connectionless time in the outage zone. Table 1 presents a comparison between the related works and the proposed scheme.

Table 1. A comparison table between the related works and the proposed scheme.

Scheme	Direction	Precaching Vehicle	Hop Count	Contribution
[10]	Same	Single	Single-hop	To minimize the outage time
[14]	Same	Single	Single-hop	To increase content delivery
[27]	Same	Multiple	Single-hop	To decrease communication outages in outage zones
[26]	Same	Multiple	Single-hop	To improve the reliability of data communication
[16]	Same	Multiple	Single-hop	To improve content precaching in outage zones
[11]	Opposite	Multiple	Single-hop	To reduce dark areas and extend RRU coverage
[28]	Opposite	Single	Single-hop	To maximize the total amount of data transferred and the average transfer rate
[12]	Both	Multiple	Single-hop	To improve outage performance via inter-RRU cooperation
Proposed scheme	Same	Multiple	Multi-hop	To minimize the connectionless time and dark area time

3. Network Model and Scheme Overview

In this section, we present the network model and provide an overview of the proposed scheme for supporting multi-hop precaching in CCVNs. Figure 1 illustrates the network model and an overview of the proposed scheme. In the network model, we assume that many vehicles are moving along roads within a vehicular network field, where multiple RSUs are deployed at regular intervals. All vehicles travel along their routes, passing several RSUs before reaching their destinations. RSUs are strategically placed by the network administrator to facilitate content downloading for vehicles within their communication range. These RSUs can communicate with one another and connect to content servers through wired backbone links. When a vehicle enters an RSU's coverage area, it sends a beacon message containing its ID, location and speed. Each RSU then updates its vehicle management table with this information, allowing it to track all vehicles within its communication range.

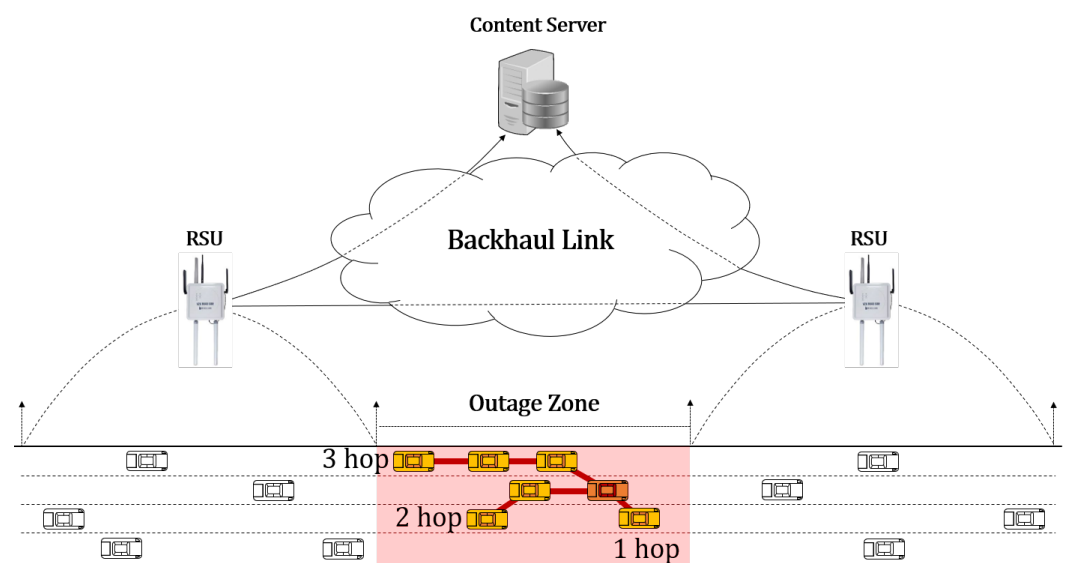


Figure 1. An overview of the proposed scheme.

In our scheme, vehicles and RSUs communicate using IEEE 802.11p, a wireless access technology in vehicular environments (WAVE) [29]. When a requester vehicle is within the communication range of an RSU, it can send a request message to the RSU for its desired content. The request message includes the content's name, size, and other relevant information. Based on this information and the RSU's transmission rate, the RSU calculates whether the requester vehicle can download all of the content while within its coverage area. However, deploying a sufficient number of RSUs along roads is challenging due to geographic constraints and high installation costs [24]. This results in vehicular networks becoming intermittently connected, known as Intermittently Connected Vehicular Networks (ICVNs) [30], because vehicles can only intermittently connect to RSUs on roads during movement. Accordingly, an outage zone can exist between two neighboring RSUs, as shown in Figure 1. Neither of the two RSUs can cover the outage zone. Thus, when the requester vehicle moves in the outage zone, it cannot connect with any RSU and download the content from the RSU [14].

If it is determined that a requester vehicle V_{req} can receive all of the intended content from an RSU RSU_j within its communication coverage, the RSU delivers all content to the requester vehicle. Otherwise, the RSU selects precaching vehicles from its vehicle management table formed through beacon messages from vehicles in its communication coverage to deliver the content in the outage zone between itself RSU_j and its next RSU RSU_{j+1} . To decide on precaching vehicles in the outage zone, the RSU first calculates the amount of content that can be delivered to V_{req} within its communication coverage.

Next, the RSU precaches the remaining content in the selected precaching vehicles' caching storage. To select the optimal one-hop precaching vehicle V_{1-hop_1} , we compare the amounts to be precached from RSU_j to V_{1-hop_1} within the communication coverage of RSU_j and delivered from V_{1-hop_1} to V_{req} in the outage zone between RSU_j and RSU_{j+1} . The amount to be delivered can be calculated by T_{R_i} , that is, the connection time between V_{req} and the i -th vehicle. Then, if the one-hop precaching vehicle V_{1-hop_1} cannot fully cover the outage zone, we select more one-hop precaching vehicles, like V_{1-hop_2} and V_{1-hop_3} , to minimize the connectionless time $t_{connectionless}$, as shown in Figure 1. In selecting additional one-hop precaching vehicles, we compare $t_{connectionless}$ with the *minimum selection time* threshold value, which is determined by the simulation results, to prevent huge control overheads due to frequent handovers with precaching vehicles. If $t_{connectionless}$ is shorter than the *minimum selection time*, additional one-hop precaching vehicles are not selected. Furthermore, to minimize the dark area time $t_{darkarea}$ during which each one-hop precaching vehicle is connected to the requester vehicle while it has no content to deliver anymore, we select precaching vehicles on additional hops. However, we restrict the number of hops in the proposed scheme, because more hops can cause excessive control overheads and reduce content delivery throughput [17]. Lastly, each of the selected precaching vehicles becomes a precaching vehicle V_{rel_k} in the connection time order with V_{req} .

The proposed scheme also addresses the scenario where multiple requester vehicles are within the coverage of an RSU, RSU_j , and each vehicle intends to utilize precaching vehicles. The overall process for handling multiple requester vehicles is similar to that for a single requester vehicle. However, when a requester vehicle requires precaching vehicles, the scheme ensures that vehicles already selected as precaching vehicles for other requester vehicles are not reused. In other words, each vehicle can only serve as a precaching vehicle for one requester vehicle at a time. This approach prevents conflicts and ensures fair resource allocation among multiple requester vehicles.

For realizing the proposed scheme in CCVNs, its computational complexity (or called time complexity) needs to be addressed in this paper. The proposed scheme consists of two algorithms, Algorithm 1 (multi-hop precaching vehicle selection) and Algorithm 2 (multiple precaching vehicle selection). First, Algorithm 1 primarily consists of one loop for the second hop and nested loops for the third hop. Assuming the selection process and capacity checks are constant times or, at most, linear, the overall computational complexity would be dominated by the third hop with a complexity of $O(n^2)$. Thus, the time complexity of Algorithm 1 is $O(n^2)$, where n is the number of vehicles being considered for selection at each hop. Second, Algorithm 2 consists of recursive calls using a full binary tree structure. Then, the full binary tree has a depth proportional to the number of vehicles, leading to multiple recursive steps. At each recursive level, it performs work with a computational complexity of $O(n)$ to select the appropriate vehicle. However, in the worst case, the recursion forms a complete binary tree, where at each level, you check n vehicles. Since the tree could have a depth of $\log n$, the total computational complexity in this case becomes $O(n \log n)$. Thus, the computational complexity of the proposed scheme is $O(n^2) + O(n \log n)$. Also, $O(n^2)$ is bigger than $O(n \log n)$. As a result, the computational complexity of the proposed scheme is $O(n^2)$, because the *Big O* notation for computational complexity leaves the part with the greatest growth and ignores the rest.

Generally, the viability of an algorithm in a vehicle network depends on various factors, such as the performance of the system, the state of the network, and the computational complexity of the algorithm. In this paper, we assume that an RSU has sufficient processing performance in terms of CPU performance, memory size, and network delay in order to perform the algorithms of the proposed method. The computational complexity of the proposed scheme is $O(n^2)$, and the computational complexity $O(n^2)$ is affected by the size of n . In the proposed scheme, n is the number of vehicles within the communication range of an RSU for selecting precaching vehicles. In general, the computational complexity $O(n^2)$ will not be significantly affected by the number of vehicles within the communication range of an RSU, because the number is not large. Therefore, we determine that the RSU can

fully execute the computational complexity $O(n^2)$ in the proposed scheme. In practice, several solutions have been proposed to have computational complexity $O(n^2)$ in vehicle networks [31,32].

We describe the details of the proposed scheme in the next Section 4.

Algorithm 1 Multi-hop precaching vehicle selection

Input: C_{req} , vehicles' mobility, $t_{out,j,req}$ and $t_{in,j+1,req}$.

Output: Linked tree of precaching vehicles.

```

1:  $C_{sum}=0$ 
2:  $C_{rest}=C_{req}$ 
3:  $hop=1$ 
4:  $V_{(1)} \leftarrow SELECTION(t_{out,j,req}, t_{in,j+1,req}, V_{req})$ 
5:  $V_{prec} \leftarrow V_{(1)}$ 
6:  $C_{(1)} \leftarrow \text{all } V_{1,k}.C_{avail} \text{ in } V_{(1)}$ 
7:  $C_{sum} = C_{sum} + C_{(1)}$ 
8: if  $C_{req} \leq C_{sum}$  then
9:   return  $V_{prec}$ 
10: end if
11:  $hop=2$ 
12: for  $V_{1,k}$  in  $V_{(1)}$  do
13:    $V_{(2^k)} \leftarrow SELECTION(V_{1,k}.t_{end}, V_{1,k}.t_{leave}, V_{1,k})$ 
14:    $V_{prec} \leftarrow V_{(2^k)}$ 
15:    $C_{(2^k)} \leftarrow \text{all } V_{2^k,l}.C_{avail} \text{ in } V_{(2^k)}$ 
16:    $C_{sum} = C_{sum} + C_{(2^k)}$ 
17:   if  $C_{req} \leq C_{sum}$  then
18:     return  $V_{prec}$ 
19:   end if
20: end for
21:  $hop=3$ 
22: for  $V_{1,k}$  in  $V_{(1)}$  do
23:   for  $V_{2^k,l}$  in  $V_{(2^k)}$  do
24:      $V_{(3^{k,l})} \leftarrow SELECTION(V_{2^k,l}.t_{end}, V_{2^k,l}.t_{leave}, V_{2^k,l})$ 
25:      $V_{prec} \leftarrow V_{(3^{k,l})}$ 
26:      $C_{(3^{k,l})} \leftarrow \text{all } V_{3^{k,l},m}.C_{avail} \text{ in } V_{(3^{k,l})}$ 
27:      $C_{sum} = C_{sum} + C_{(3^{k,l})}$ 
28:     if  $C_{req} \leq C_{sum}$  then
29:       return  $V_{prec}$ 
30:     end if
31:   end for
32: end for
33: return  $V_{prec}$ 

```

Algorithm 2 Multiple precaching vehicle selection

Input: hop , C_{rest} , t_{ct1} and t_{ct2} .

Output: Linked tree of precaching vehicles.

Function: $SELECTION(t_{ct1}, t_{ct2}, V_{src})$:

```

1: if  $t_{ct2} - t_{ct1} < t_{ms}$  then
2:   return  $NULL$ 
3: end if
4: if  $hop < 3$  then
5:    $V_{select} \leftarrow V_i$  with  $\max t_{conn}(V_i, V_{src})$  between  $t_{ct1}$  and  $t_{ct2}$ 
6: else
7:    $V_{select} \leftarrow V_i$  with  $\max C_{avail}(V_i, V_{src})$  between  $t_{ct1}$  and  $t_{ct2}$ 
8: end if

```

Algorithm 2 Cont.

```

9:  $C_{rest} = C_{rest} - V_{select} \cdot C_{avail}$ 
10: if  $C_{rest} \leq 0$  then
11:   return  $V_{select}$ 
12: end if
13: if  $t_{ms} \leq V_{select} \cdot t_{meet} - t_{ct1}$  then
14:    $V_{select}[Left] \leftarrow SELECTION(t_{ct1}, V_{select} \cdot t_{meet}, V_{src})$ 
15: end if
16: if  $C_{rest} \leq 0$  then
17:   return  $V_{select}$ 
18: end if
19: if  $t_{ms} \leq t_{ct2} - V_{select} \cdot t_{leave}$  then
20:    $V_{select}[Right] \leftarrow SELECTION(V_{select} \cdot t_{leave}, t_{ct2}, V_{src})$ 
21: end if
22: return  $V_{select}$ 

```

4. The Proposed Scheme

In this section, we describe the proposed scheme that selects multi-hop multiple precaching vehicles to perform precaching for a requester vehicle in outage zones of CCVNs. First, we describe the decision on the necessity of precaching to provide the requested content to the requester vehicle in Section 4.1. Then, we describe the *minimum selection time* required for the selection of the precaching vehicle to cover the connectionless time in Section 4.2. According to the precaching decision and the *minimum selection time*, we describe the selection scheme of the first-hop multiple precaching vehicles in Section 4.3. Subsequently, we describe the selection scheme for multiple intermediary precaching vehicles in Section 4.4. Lastly, we describe the selection scheme for the last hop's multiple precaching vehicles in Section 4.5.

4.1. Checking the Necessity for Precaching

When a vehicle V_{req} moves within the coverage of an RSU R_j and wants to download intended content, it sends a request message for the intended content to the RSU. After receiving the request message, the RSU obtains the information about the requested content from the content server, which contains the content's ID and total size C_{req} . To decide on precaching, the RSU should compare the total size of the requested content with the available amount of content that can be provided to the requester vehicle V_{req} within its coverage area until the requester vehicle is out of the coverage area of the RSU. To know the available amount of content, we first have to calculate the dwell time $t_{dwell,j,req}$ of V_{req} within R_j 's coverage area as follows:

$$t_{dwell,j,req} = \frac{x_j + Range_j - x_{req}}{v_{req}}, \quad (1)$$

where x_j is the location of R_j , $Range_j$ is the communication range of R_j , x_{req} is the location of V_{req} , and v_{req} is the velocity of V_{req} . Then, based on the dwell time, we can denote the downloadable amount $C_{down,j,req}$ in which it is possible for V_{req} to be provided from R_j during $t_{dwell,j,req}$ as follows:

$$C_{down,j,req} = t_{dwell,j,req} \times r_j, \quad (2)$$

where r_j is the transmission rate of R_j . Based on $C_{down,j,req}$, R_j can determine the remaining amount $C_{rest,j,req}$ of the content that has to be provided to V_{req} within the outage zone as follows:

$$C_{rest,j,req} = C_{req} - C_{down,j,req}. \quad (3)$$

If $C_{rest,j,req}$ is smaller than 0, the requested content does not need to be precached. Otherwise, R_j has to precache the content for V_{req} . Because there is a limit to the amount of content that V_{req} can receive from precaching vehicles due to the mobility of V_{req} and the coverage area of RSUs, the whole of $C_{rest,j,req}$ cannot be provided within the outage zone. Therefore, R_j has to determine the duration of time $t_{out,j,req}$ during which V_{req} remains within the outage zone as follows:

$$t_{out,j,req} = \frac{(x_{j+1} - Range_{j+1}) - (x_j + Range_j)}{v_{req}}, \quad (4)$$

where $Range_{j+1}$ is the communication range of R_{j+1} and x_{j+1} is the location of R_{j+1} . Then, using $t_{out,j,req}$, the RSU can estimate the available amount $C_{out,j,req}$ of content within the outage zone for V_{req} as follows:

$$C_{out,j,req} = t_{out,j,req} \times r_{V2V}, \quad (5)$$

where r_{V2V} is the transmission rate of the vehicle-to-vehicle communication. Consequently, the RSU can define the amount $C_{need,req}$ of the content that should be precached from the outage zone as follows:

$$C_{need,req} = \begin{cases} C_{rest,j,req} & \text{if } C_{rest,j,req} < C_{out,j,req} \\ C_{out,j,req} & \text{else.} \end{cases} \quad (6)$$

To provide the $C_{need,req}$ of the requested content within the outage zone, the RSU has to select multiple precaching vehicles. The selected precaching vehicles precache the assigned part of the requested content until V_{req} leaves this RSU's coverage area and will deliver the precached content to V_{req} within the outage zone.

4.2. The Minimum Selection Time

For a requester vehicle to receive content from a precaching vehicle in an outage zone, the requester vehicle must first create a connection with the precaching vehicle. In addition, when the content delivery from this precaching vehicle is completed, the requester vehicle must terminate the connection with it to receive content from a new precaching vehicle. After that, the requester vehicle can create a new connection with the new precaching vehicle to receive content from it. This creation and termination process for the connection between a requester vehicle and a precaching vehicle causes a control overhead that requires multiple packet transmissions. Moreover, this process leads to a larger control overhead for the connection based on multi-hop communications. In the proposed scheme, there are two times, the connectionless time and the dark area time, during which a requester vehicle cannot receive content in an outage zone. Accordingly, it is an important issue to minimize the connectionless time and the dark area time for a requester vehicle in the outage zone while using a small number of precaching vehicles. Thus, when an RSU selects precaching vehicles, the proposed scheme needs to consider the *minimum selection time* to prevent frequent handovers between a requester vehicle and precaching vehicles. Based on the connections of first-hop precaching vehicles, the requester vehicle can obtain the content from them. The connectionless time is the *minimum selection time* between the connection time of the $(n - 1)$ -th precaching vehicle and the connection time of the n -th precaching vehicle. In that case, the RSU selects more precaching vehicles to cover the connection time. Then, since they have a limited time to download the content within the coverage area of the RSU, they may have insufficient content to provide to the requester vehicle during their connection times when the connection time is longer than the time required to deliver it. The dark area time is the time during which the requester vehicle cannot receive the content from the lower-hop precaching vehicle due to its limited time to deliver the content. In that case, the RSU selects additional-hop precaching vehicles to cover the dark area time.

In the proposed scheme, when an RSU selects additional precaching vehicles to cover the connectionless time and the dark area time in an outage zone for a requester vehicle, it can cover even a very short connectionless time and dark area time by utilizing an excessively large number of precaching vehicles. However, the large number of precaching vehicles causes frequent handovers to the requester vehicle in the outage zone. Therefore, we set the *minimum selection time* t_{ms} based on simulations to prevent the selection by including too many precaching vehicles while not degrading the performance of the proposed scheme in terms of the connectionless time and the dark area time. This means that the RSU does not select more precaching vehicles to cover too short a connection time and dark area time when they are less than the *minimum selection time*. Figures 2 and 3 show the simulation results for the performance of the proposed scheme for different *minimum selection times* in terms of the connectionless time and the dark area time, respectively.

As shown in Figures 2 and 3, a shorter *minimum selection time* leads to better performance of the proposed scheme in terms of the connectionless time and the dark area time. However, for times of less than 3 seconds, the performances converge. For this reason, we use a 3 s *minimum selection time* in the proposed scheme.

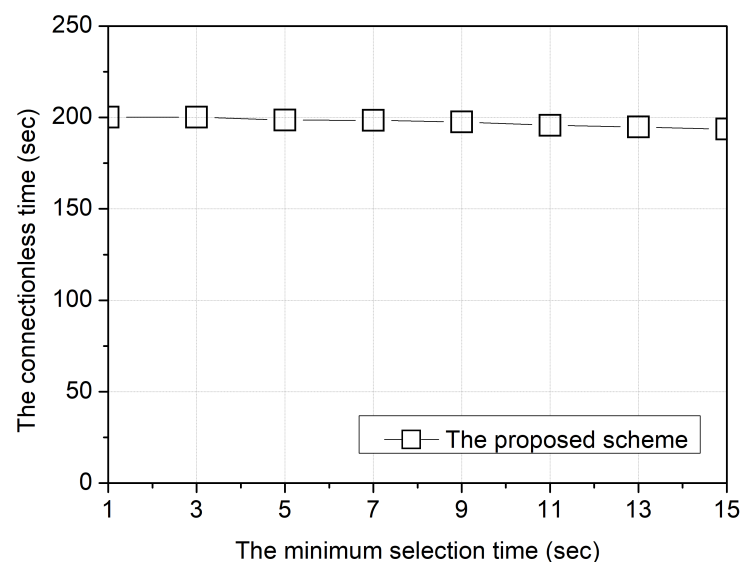


Figure 2. The connectionless time for the minimum selection time.

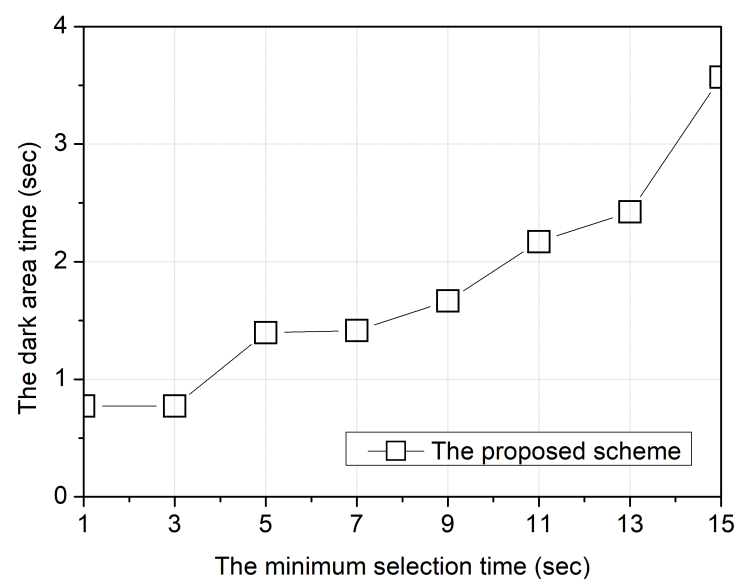


Figure 3. The dark area time for the minimum selection time.

4.3. Multiple First-Hop Precaching Vehicle Selection

The first-hop precaching vehicles are first selected because the first hop is the foundation of connection for the requester vehicle in multi-hop communication, as shown in Figure 4. Since the requester vehicle is served by multi-hop precaching vehicles, the selection of the first-hop precaching vehicle is more likely to focus on the connection time to the requester vehicle. However, as the requester vehicle has to access RSUs to download content within their coverage area because they have sufficient communication resources in the content provision services, the connection time between the requester vehicle and a precaching vehicle must be considered based on the connectionless time within the outage zone. To obtain the connectionless time, the RSU has to calculate two times: the leave time from the current RSU of the requester vehicle, and the entrance time to the next RSU of the requester vehicle. The RSU can determine the leave time based on $t_{dwell,j,req}$. Then, it can calculate the entrance time $t_{enter,j+1,req}$ to the next RSU R_{j+1} of V_{req} as follows:

$$t_{enter,j+1,req} = \frac{(x_{j+1} - Range_{j+1} - x_{req})}{v_{req}}. \quad (7)$$

Based on $t_{enter,j+1,req}$ and $t_{out,j,req}$, the RSU selects a first-hop precaching vehicle as shown in Figure 4. Then, the RSU can obtain the meet time and leave time between the requester vehicle and a first-hop precaching vehicle. For example, when representing two vehicles as V_a and V_b , we can obtain the meet time $t_m(a, b)$ and the leave time $t_l(a, b)$, respectively, as follows:

$$t_m(a, b) = \frac{x_b - x_a + Range_a}{v_a - v_b}, \quad (8)$$

and

$$t_l(a, b) = \frac{x_b - x_a - Range_a}{v_a - v_b}, \quad (9)$$

where x_a and x_b are the location of the two vehicles; v_a and v_b represent the velocity of them; and $Range_a$ is the communication range of V_a .

In the same way, the RSU can calculate the meet time and leave time between V_{req} and a candidate vehicle for the i -th precaching vehicle V_i within the outage zone using $t_{dwell,j,req}$ and $t_{enter,j+1,req}$, respectively, as follows:

$$t_{meet,req,i}(t_{dwell,j,req}, t_{enter,j+1,req}) = \begin{cases} t_{dwell,j,req} & \text{if } t_m(req, i) < t_{dwell,j,req} \\ & t_l(req, i) < t_{dwell,j,req} \\ t_{enter,j+1,req} & \text{if } t_m(req, i) > t_{dwell,j,req} \\ & t_l(req, i) > t_{dwell,j,req} \\ t_m(req, i) & \text{if } t_m(req, i) < t_l(req, i) \\ t_l(req, i) & \text{else,} \end{cases} \quad (10)$$

and

$$t_{leave,req,i}(t_{dwell,j,req}, t_{enter,j+1,req}) = \begin{cases} t_{dwell,j,req} & \text{if } t_m(req, i) < t_{dwell,j,req} \\ & t_l(req, i) < t_{dwell,j,req} \\ t_{enter,j+1,req} & \text{if } t_m(req, i) > t_{dwell,j,req} \\ & t_l(req, i) > t_{dwell,j,req} \\ t_m(req, i) & \text{if } t_m(req, i) > t_l(req, i) \\ t_l(req, i) & \text{else.} \end{cases} \quad (11)$$

With the two calculated times, the RSU can obtain the connection time $t_{conn,req,i}$ with $t_{dwell,j,req}$ and $t_{enter,j+1,req}$ between V_{req} and V_i within the outage zone as follows:

$$t_{conn,req,i}(t_{dwell,j,req}, t_{enter,j+1,req}) = t_{leave,req,i}(t_{dwell,j,req}, t_{enter,j+1,req}) - t_{meet,req,i}(t_{dwell,j,req}, t_{enter,j+1,req}). \quad (12)$$

Based on the $t_{conn,req,i}$ of every V_i , the V_i with the biggest connection time is selected as the first-hop precaching vehicle $V_{1.1}$ and its available amount $C_{avail,req,1.1}$ of content is calculated as follows:

$$C_{avail,req,1.1} = t_{conn,req,1.1} \times r_{1.1}, \quad (13)$$

where $r_{1.1}$ is the transmission rate of $V_{1.1}$. When $C_{need,req}$ is smaller than $C_{avail,req,1.1}$, R_j finishes the selection process. Otherwise, it keeps selecting another first-hop precaching vehicle. The RSU compares the connectionless times before and after the connecting of V_{req} and V_i with t_{ms} , respectively. The connectionless time can be divided into before and after based on the connection time with $V_{1.1}$. Then, the RSU checks whether more precaching vehicles are needed to cover the outage zone by comparing t_{ms} and the two connectionless times. If the connectionless time is longer than t_{ms} , the RSU searches for more precaching vehicles to cover the outage zone. Until all connectionless times are shorter than t_{ms} , R_j searches more first-hop k -th precaching vehicles within the connectionless area based on the biggest connection time $t_{conn,req,i}(t_{leave,req,1.(k-1)}, t_{meet,req,1.k+1})$, where $t_{leave,req,1.(k-1)}$ denotes the disconnect time between V_{req} and $V_{1.(k-1)}$, and $t_{meet,req,1.k+1}$ denotes the meeting time between V_{req} and $V_{1.k+1}$. After selecting the new precaching vehicle, R_j reorders the number of precaching vehicles in the order in which they meet V_{req} . When $C_{need,req} \leq \sum_{k=1}^K C_{avail,req,1.k}$ and all connectionless times are shorter than t_{ms} , R_j stops finding other first-hop precaching vehicles.

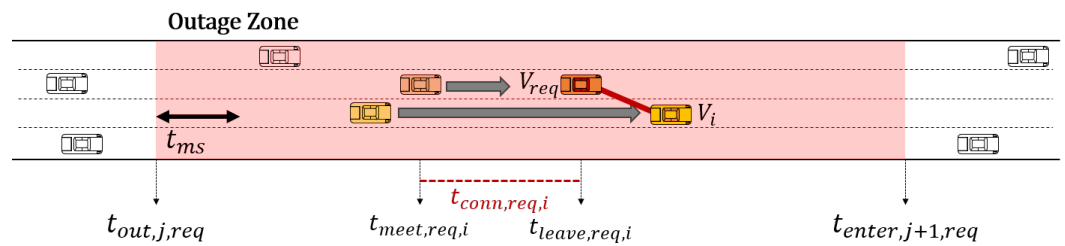


Figure 4. The process of first-hop precaching vehicle selection.

4.4. Multiple Second-Hop Precaching Vehicle Selection

The RSU has to search the second-hop precaching vehicles when the first-hop precaching vehicles have too long a connection time to V_{req} by comparing this time with the time required to deliver the content, as shown in Figure 5. Since a first-hop precaching vehicle $V_{1.k}$ has a limited time to deliver the content due to the dwell time of V_{req} , it cannot deliver all parts of its $C_{avail,req,1.k}$. Also, when $V_{1.k}$ has a shorter time to deliver than the connection time to V_{req} , it cannot deliver more content to V_{req} in the outage zone. Therefore, the RSU should know the precaching time $t_{prec,1.k}$ of $V_{1.k}$ and $t_{prec,1.k}$, which can be determined as follows:

$$t_{prec,1.k} = \begin{cases} t_{dwell,j,1.k} & \text{if } t_{dwell,j,1.k} < t_{dwell,j,req} \\ t_{dwell,j,req} & \text{else.} \end{cases} \quad (14)$$

Based on $t_{prec,1.k}$, the RSU can define the precaching amount $C_{prec,1.k}$ of $V_{1.k}$ as follows:

$$C_{prec,1.k} = t_{prec,1.k} \times r_j. \quad (15)$$

In this second-hop precaching vehicle selection, the RSU has to consider t_{ms} . If $t_{conn,req,1.1} - t_{prec,1.k}$ is longer than t_{ms} within the outage zone, $V_{1.k}$ needs to connect second-hop precaching vehicles within its connection time after it delivers all of its precached content to V_{req} in the outage zone. Then, a second-hop precaching vehicle has to deliver its

precached content after $V_{1,k}$ delivers $C_{prec,1,k}$. Therefore, the RSU has to define the end time $t_{end,1,k}$ to precache $V_{1,k}$'s $C_{prec,1,k}$ as follows:

$$t_{end,1,k} = t_{meet,req,1,k} + \frac{C_{prec,1,k}}{r_{1,k}}, \quad (16)$$

where $r_{1,k}$ is the transmission rate of $V_{1,k}$. Based on $t_{end,1,k}$ and $t_{leave,req,i}(t_{dwell,j,req}, t_{enter,j+1,req})$, the RSU can select the second-hop precaching vehicle from the time when $V_{1,k}$ finishes delivering its precached content to V_{req} to the time when $V_{1,k}$ and V_{req} are disconnected by communication range limitation, as shown in Figure 5. The connection time of the second-hop candidate vehicle is denoted by $t_{conn,req,2^k,l}(t_{end,1,k}, t_{leave,req,1,k})$. Then, the candidate vehicle with the maximum connection time becomes the second-hop precaching vehicle within the connection time between V_{req} and $V_{1,k}$, which is denoted as $V_{2^k,l}$. Then, the connection time of the second-hop l -th precaching vehicle must be inside the period between $t_{leave,req,2^k,(l-1)}$ and $t_{meet,req,2^k,l}$. Also, in every selection of the second-hop precaching vehicles, the RSU has to consider t_{ms} .

After the selection of second-hop precaching vehicles, they precache the assigned amount of the content, which is $C_{prec,2^k,l}$. Then, when the first-hop precaching vehicle finishes its precached content delivery in the outage zone and it has more time in which it is possible to deliver the content during its connection time within the outage zone, the second-hop precaching vehicle delivers its precached content to the first-hop precaching vehicle. The first-hop precaching vehicle delivers the content to V_{req} , as shown in Figure 5.

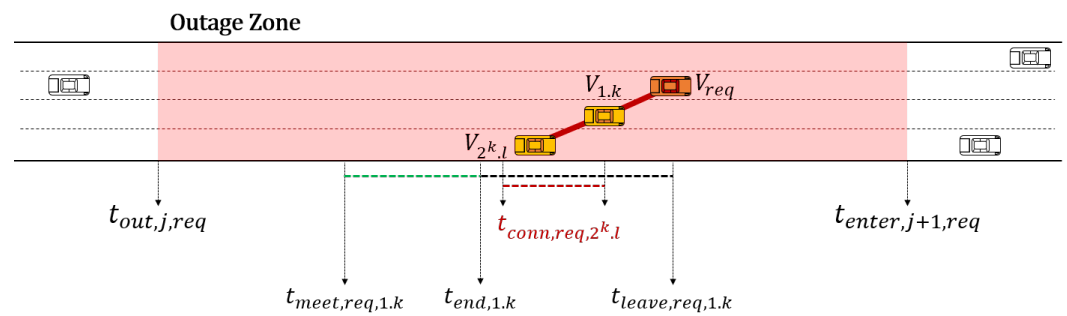


Figure 5. The process of second-hop precaching vehicle selection.

4.5. Multiple Third-Hop Precaching Vehicle Selection

In the proposed scheme, because the second-hop precaching vehicles may have insufficient precaching time, as shown in Figure 6, the RSU needs to select third-hop precaching vehicles for the requester vehicle in the outage zone. This is necessary to ensure that the requester vehicle V_{req} receives the requested content without interruption during its journey through the outage zone. The selection of third-hop precaching vehicles addresses the limitations of the second-hop precaching vehicles by providing additional precached content when the second-hop precaching vehicles cannot cover the entire duration required for content delivery. In the selection of the third-hop precaching vehicle, the RSU considers $t_{end,2^k,l}$ based on $t_{prec,2^k,l}$. Then, using $t_{leave,1,k,2^k,l}$, the RSU can define the dark area time of $V_{2^k,l}$. If the dark area time is longer than t_{ms} , it searches third-hop precaching vehicles. It is necessary to consider the connection time of the vehicles with $V_{2^k,l}$ within the range between $t_{end,2^k,l}$ and $t_{leave,1,k,2^k,l}$ to guarantee the effective precaching of content. This range ensures that the third-hop precaching vehicles can connect to $V_{2^k,l}$ right after $V_{2^k,l}$ finishes its content transmission and before it loses connection with $V_{1,k}$. Consequently, it guarantees that the third-hop precaching vehicle can continue the content transmission without interruption, maintaining the integrity and continuity of the content precaching process.

After the third-hop precaching vehicle precaches all of its precached content, it cannot be helped by other vehicles. Therefore, the RSU should consider its precaching time in

this third-hop precaching vehicle selection. Thus, the available amount of $C_{avail,req,3^{k,l}.m}$ of a third-hop m -th precaching vehicle $V_{3^{k,l}.m}$ is denoted as follows:

$$C_{avail,req,3^{k,l}.m} = \begin{cases} t_{conn,req,3^{k,l}.m} \times r_{3^{k,l}.m} & \text{if } t_{conn,req,3^{k,l}.m} < t_{prec,3^{k,l}.m} \\ t_{prec,3^{k,l}.m} \times r_{3^{k,l}.m} & \text{else.} \end{cases} \quad (17)$$

The RSU selects the vehicle with the biggest $C_{avail,req,3^{k,l}.m}$ as the third-hop precaching vehicle between $t_{end,2^{k,l}}$ and $t_{leave,1,k,2^{k,l}}$, as shown in Figure 6. The connection time of $V_{3^{k,l}.m}$ must be inside the period between $t_{leave,2^{k,l},3^{(k-1),l}.m}$ and $t_{meet,2^{k,l},3^{(k+1),l}.m}$. This selection also is conducted based on t_{ms} .

All precaching vehicles download their assigned parts of content from the RSU until V_{req} leaves the RSU. When the first-hop precaching vehicle meets V_{req} , it delivers its precached content to V_{req} in the outage zone. During the remaining connection time after providing all its precached content, the second-hop precaching vehicle delivers its precached content to the first-hop precaching vehicle when they meet. Then, the first-hop precaching vehicle precaches that content to V_{req} . If the second-hop precaching vehicle is still connected to the first-hop precaching vehicle after delivering all its precached content, the third-hop precaching vehicle delivers its precached content to the second-hop precaching vehicle when they meet. The second-hop precaching vehicle precaches the content to the first-hop precaching vehicle and the first-hop precaching vehicle precaches the content to V_{req} , as shown in Figure 6.

The RSU first selects the first-hop precaching vehicles, as shown in Algorithm 1. If the available number of them is enough, the RSU finishes this process. Otherwise, the RSU selects the second-hop precaching vehicles based on the first-hop precaching vehicles. Every time a second-hop precaching vehicle for each first-hop precaching vehicle is selected, the RSU checks the available number of them. If there are not enough second-hop precaching vehicles, the RSU selects the third-hop precaching vehicles based on previous-hop precaching vehicles. In this selection, the RSU also checks the available number of the selected vehicles as a precaching vehicle in every selection.

The multiple precaching vehicle selection is conducted within the given range, as shown in Algorithm 2. In the case of first-hop precaching vehicle selection, the given range is the connectionless time between $t_{out,j,req}$ and $t_{enter,j+1,req}$. Based on the algorithm, the RSU first checks that the connectionless time is longer than t_{ms} . Then, except the third-hop precaching vehicle selection, the RSU selects the vehicle with the longest connection time with V_{req} . In the third-hop precaching vehicle selection, the RSU considers the available number of candidate vehicles. If the selected vehicle can resolve the requested size of the content from V_{req} , the RSU finishes this process. Otherwise, the RSU selects more relaying vehicles by comparing the rest time to t_{ms} . Since the early-connected precaching vehicle has a higher priority than the later-connected precaching vehicle, $V_{select}[Left]$ is selected before $V_{select}[Right]$.

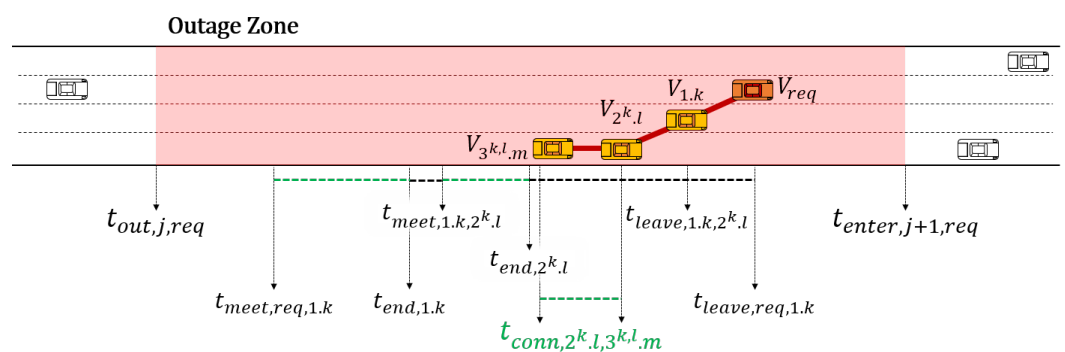


Figure 6. The process of third-hop precaching vehicle selection.

5. Performance Evaluation

In this section, we evaluate the performance of the proposed scheme through simulations. First, we describe the simulation environment and the performance evaluation metrics used. Next, we compare the performance of the proposed scheme with that of existing schemes based on the simulation results.

5.1. Simulation Environments and Performance Evaluation Metrics

For the performance evaluation of the proposed scheme, we conducted simulations using Network Simulator 3 (NS3) [33]. We implemented an enhanced Manhattan mobility model to reflect an urban environment suitable for CCVNs. Table 2 shows the simulation parameters used in our simulations. We focus on how much of the outage zone can be covered according to the physical properties of the vehicle and the RSU. We evaluated the proposed scheme in a road environment where RSUs are located 8 km away and the density of the vehicles is 100 per km². We used Gaussian distribution to create a road situation in which vehicles move at a constant speed with an error range of 10% at each average speed. The average speed is 40 km/h, and vehicles on the road have speeds between 20 km/h and 120 km/h depending on the Gaussian distribution. Among these vehicles, the vehicles within the communication range of an RSU send their location and speed information to the RSU using beacon messages. All entities, including RSUs and vehicles, communicate using 802.11p WAVE and have a maximum transmission rate of 54 Mbps. RSUs and vehicles have a maximum communication range of 800 and 200 meters, respectively. Since RSUs are deployed every 8 km, the outage zone is 6.4 km. The requester vehicle requests 2000 MB of content in such a road environment. The *minimum selection time* is set to 3 s. If the connectionless time or the dark area time exceeds 3 s, the precaching vehicle is selected in the proposed scheme.

Table 2. Simulation environment parameters.

Parameters	Value
The radio channel rate	6 Mbps
The distance between RSUs	8 km
The outage zone	6.4 km
The vehicle density	100 per km ²
The request size of the content	2000 MB
The vehicle speed	[20, 120] km/h
The V2I communication range	800 m
The V2V communication range	200 m
The storage limit of a vehicle	512 GB
The storage limit of an RSU	1 TB
The amount of content	1,000,000
The prediction error	[0,1]
The minimum selection time	3 s

We evaluated the performance of the proposed scheme by comparing it with two existing schemes, the Adaptive Carry–Store–Forward (ACSF) scheme [10] and the Set Ranking-based Precaching (SRP) scheme [16], both of which rely on single-hop communication for selecting precaching vehicles. ACSF uses only one precaching vehicle moving in the same direction as the requester vehicle. In ACSF, when a requester vehicle reaches the midpoint of an RSU’s communication range, vehicles driving in the same direction behind the requester vehicle become candidates for precaching. The RSU calculates the time when these candidate vehicles are connected to both the RSU and the requester vehicle. Based on these calculations, only the vehicle that can deliver the most content to the requester vehicle is selected as the precaching vehicle for the outage zone. On the other hand, SRP efficiently selects multiple vehicles for precaching to deliver more content in outage zones. SRP identifies sets of vehicles capable of successful content precaching by calculating download and

delivery times and content amounts based on the vehicles' mobility information. It then ranks each set according to the number of vehicles and their available resources. Finally, SRP selects the highest-ranked set and utilizes those vehicles as precaching vehicles for content delivery in the outage zone.

As performance evaluation metrics, we measured the connectionless time and dark area time under various simulation conditions. Connectionless time refers to the period when the requester vehicle remains in the outage zone without being connected to any precaching vehicle. Dark area time refers to the time during which the requester vehicle is connected to a precaching vehicle but cannot receive any content because the precaching vehicle has no more content to deliver.

5.2. Simulation Results

Figure 7a shows the dark area time relative to vehicle density. As vehicle density increases, more candidate vehicles become available, which reduces the dark area time for all schemes. ACSF exhibits the highest dark area time across all densities. Although dark area time decreases with higher vehicle density, it remains elevated because often, only one precaching vehicle is available, leading to insufficient content delivery. SRP reduces dark area time compared to ACSF by utilizing multiple precaching vehicles; however, since it operates within a single-hop range, there can still be instances of insufficient content during connection times. Overall, the dark area time is lower than that of ACSF due to connections with other vehicles. The proposed scheme demonstrates the lowest dark area time by employing multiple precaching vehicles in a multi-hop range, facilitating continuous content delivery from RSUs. As vehicle density increases, dark area time further decreases, indicating the most efficient performance.

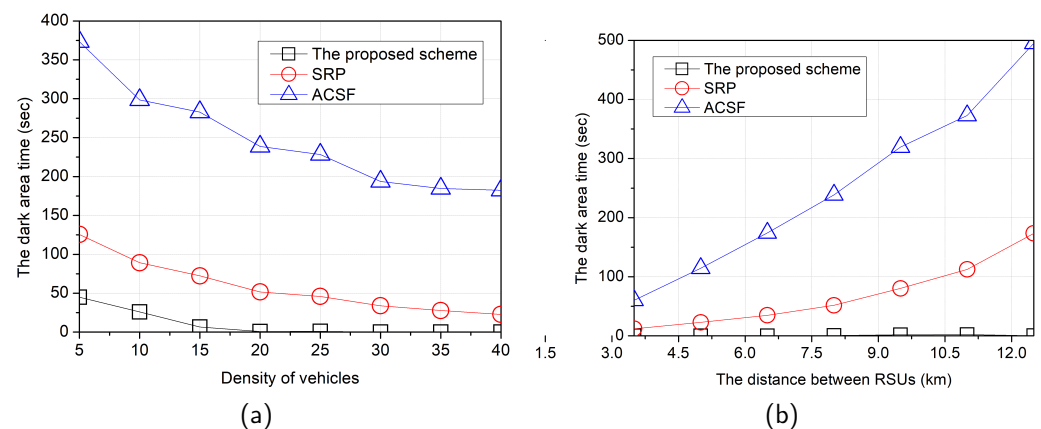


Figure 7. The dark area time according to (a) the density of vehicles; (b) the distance between RSUs.

Figure 7b shows the dark area time relative to the distance between RSUs. As the distance between RSUs increases, the dark area time also rises because vehicles spend more time in dark areas without receiving content. ACSF, which uses a single precaching vehicle, often fails to deliver all content within the connection time, leading to a significant increase in dark area time as the RSU distance increases. Although SRP utilizes multiple precaching vehicles, its single-hop communication range limits the number of candidate vehicles, causing the dark area time to rise rapidly with increasing RSU distance. In contrast, the proposed scheme employs multiple precaching vehicles within a multi-hop communication range, effectively delivering content during connection times. This approach minimizes the dark area time, even as the RSU distance increases. Overall, the proposed scheme consistently achieves the shortest dark area time, with only a moderate increase as the distance between RSUs grows.

Figure 8a shows the connectionless time for the density of vehicles. As vehicle density increases, the connectionless time decreases due to the increase in available options for precaching vehicles. ACSF shows the highest connectionless time across all densities. Since

it uses a single precaching vehicle in a single-hop communication range, it often lacks sufficient candidate vehicles for data delivery, resulting in longer dark area times. SRP shows a lower connectionless time than ACSF. It uses multiple precaching vehicles, providing more candidate vehicles and reducing connectionless time. However, it still operates within a single-hop communication range, leading to higher connectionless time compared to the proposed scheme. The proposed scheme shows the lowest overall connectionless time densities. Utilizing multiple precaching vehicles in a multi-hop communication range maximizes the number of candidate vehicles and minimizes dark area time, offering the most efficient performance.

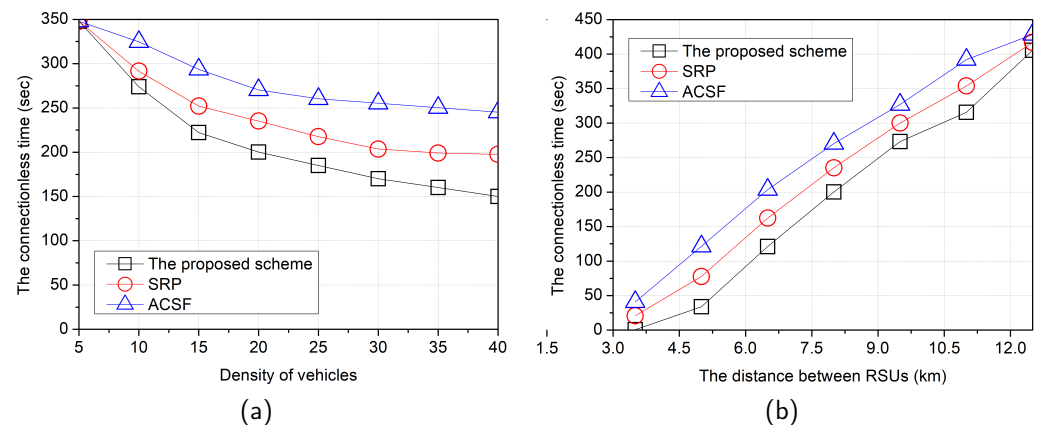


Figure 8. The connectionless time according to (a) the density of vehicles; (b) the distance between RSUs.

Figure 8b shows the connectionless time in relation to vehicle density. As vehicle density increases, connectionless time decreases due to the greater availability of precaching vehicles. ACSF exhibits the highest connectionless time across all densities. Since it relies on a single precaching vehicle within a single-hop communication range, it often lacks sufficient candidate vehicles for data delivery, resulting in longer connectionless times. In contrast, SRP demonstrates a lower connectionless time than ACSF by utilizing multiple precaching vehicles, which increases the number of candidates and reduces connectionless time. However, it still operates within a single-hop communication range, resulting in higher connectionless time compared to the proposed scheme. The proposed scheme consistently achieves the lowest connectionless time across all densities. By employing multiple precaching vehicles within a multi-hop communication range, it maximizes the number of candidate vehicles and minimizes connectionless time, thus providing the most efficient performance.

Figure 9a shows the dark area time in relation to the data rate. While increasing the data rate enables more data to be transmitted during connection time, it does not significantly impact dark area time due to the finite amount of data and the limited number of precaching vehicles. ACSF exhibits a high dark area time of approximately 250 s across various data rates, as it relies on a single precaching vehicle, resulting in the highest dark area time. In comparison, SRP maintains a constant dark area time of about 50 s, which is lower than that of ACSF. By utilizing multiple precaching vehicles, SRP reduces the dark area time relative to ACSF, although it remains constrained by the single-hop communication range. The proposed scheme achieves the lowest dark area time by employing multiple precaching vehicles within a multi-hop communication range. This efficient precaching strategy effectively minimizes dark area time, regardless of the data rate.

Figure 9b shows the dark area time in relation to the size of the requested content. As the content size increases, more precaching vehicles are required, leading to larger dark areas and increased dark area time. ACSF relies on a single precaching vehicle, resulting in a constant dark area time regardless of content size due to its limited precaching capacity.

In contrast, SRP employs multiple relaying vehicles within a single-hop communication range. However, the insufficient number of candidate vehicles causes the dark area time to rise sharply as content sizes increase. The proposed scheme utilizes multiple precaching vehicles within a multi-hop communication range, enabling continuous content delivery even in dark areas. Consequently, dark area time remains constant, irrespective of content size.

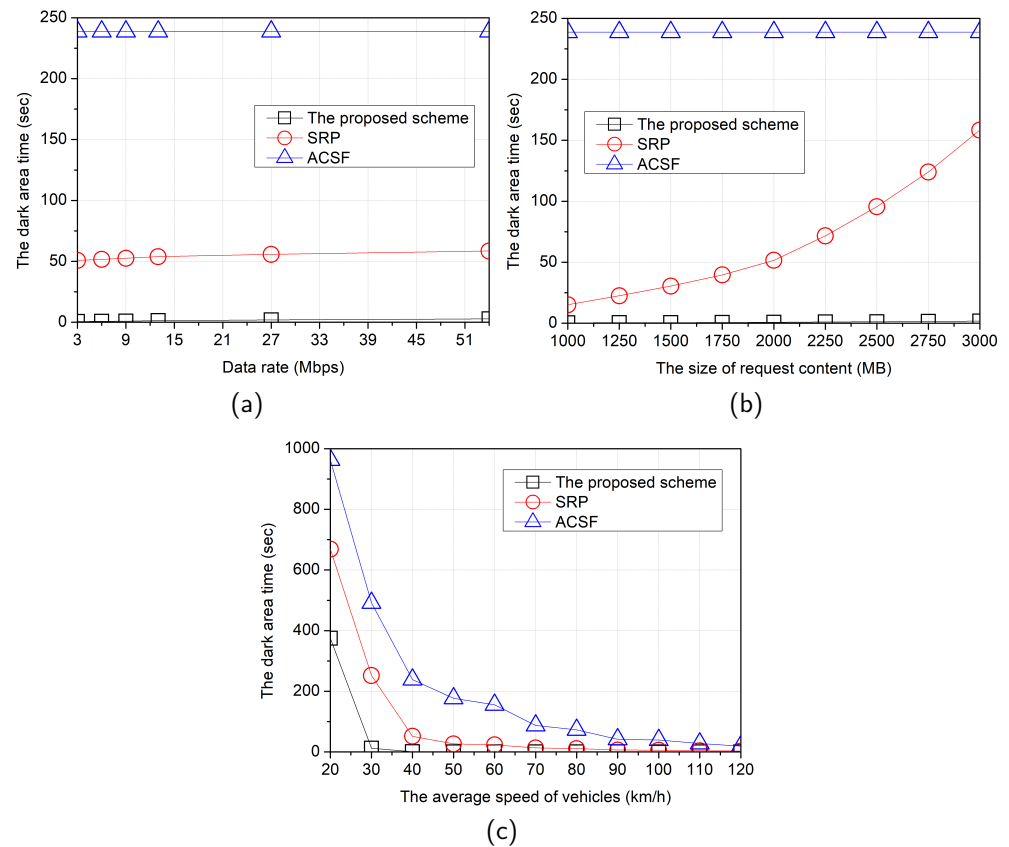


Figure 9. The dark area time according to (a) the data rate; (b) the size of the requested content; (c) the average speed of vehicles.

Figure 9c shows the dark area time as a function of the average speed of vehicles. As vehicle speed increases, connection times with RSUs and relay vehicles shorten, leading to a decrease in dark area time since vehicles pass through dark areas more quickly. ACSF exhibits a high dark area time at 20 km/h; although it decreases with higher speeds, it remains the highest compared to the other schemes. SRP also shows elevated dark area time at speeds of 40 km/h or less, but it decreases sharply with increasing speed, approaching zero at speeds of 50 km/h or more. The proposed scheme demonstrates a high dark area time below 30 km/h, but this falls to nearly zero at speeds above 30 km/h, indicating the most efficient performance at higher speeds.

Figure 10a shows the connectionless time as a function of data rate. Initially, as the data rate increases, the connectionless time rises due to insufficient content delivery time, but stabilizes beyond a data rate of 15. ACSF employs a single precaching vehicle within a single-hop communication range, leading to a decrease in connectionless time at higher data rates, although it remains elevated at lower data rates compared to the other schemes. SRP utilizes multiple precaching vehicles, resulting in lower connectionless time than ACSF. Although connectionless time decreases with higher data rates, it remains higher than that of the proposed scheme due to its single-hop communication limitations. The proposed scheme experiences a noticeable increase in connectionless time at low data rates; however, beyond a data rate of 15, it effectively uses multiple precaching vehicles in a multi-hop communication range, achieving the lowest connectionless time among all schemes.

Figure 10b shows the connectionless time as a function of the size of the requested content. As the size of the requested content increases, more precaching vehicles are required, leading to potential dark areas and an increase in connectionless time. ACSF exhibits a constant connectionless time regardless of the size of the requested content because it relies solely on a single precaching vehicle. In contrast, SRP employs multiple precaching vehicles within a single-hop range. As the size of the requested content increases, the limited number of available precaching vehicles results in a significant rise in connectionless time. Although connectionless time does increase with content size, the proposed scheme maintains a relatively lower connectionless time compared to both SRP and ACSF. This scheme effectively delivers larger amounts of content through multiple precaching vehicles in a multi-hop communication range.

Figure 10c shows the connectionless time as a function of the average speed of vehicles. Faster vehicle speeds shorten connection times with RSUs and precaching vehicles, thereby reducing the connectionless time as vehicles quickly pass through dark areas. ACSF exhibits higher connectionless time than the other schemes; however, it decreases as vehicle speed increases due to its reliance on a single precaching vehicle. In contrast, SRP employs multiple precaching vehicles, resulting in lower connectionless time compared to ACSF. Nevertheless, its connectionless time remains higher than that of the proposed scheme due to its single-hop communication approach. The proposed scheme utilizes multiple precaching vehicles in a multi-hop communication range, ensuring stable content delivery and achieving the lowest connectionless time across varying vehicle speeds.

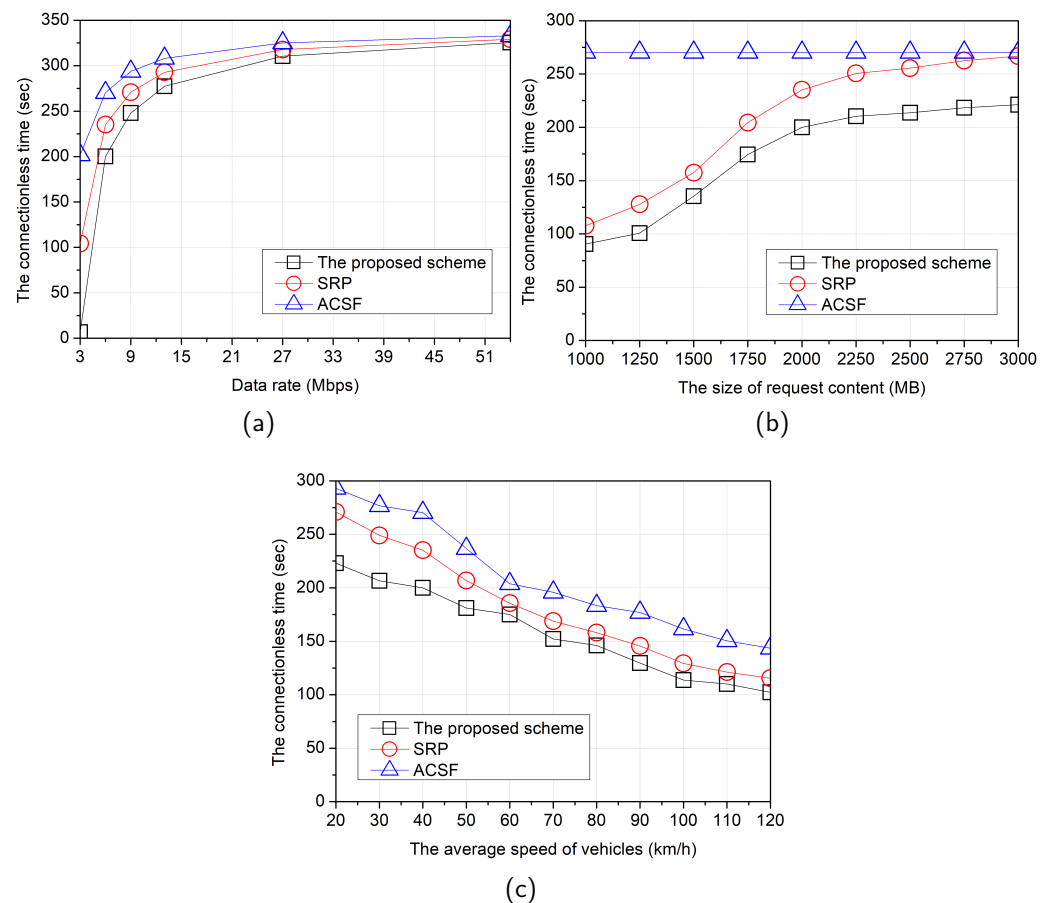


Figure 10. The connectionless time according to (a) the data rate; (b) the size of the requested content; (c) the average speed of vehicles.

6. Conclusions

With the development of self-driving cars, mobile data traffic has surged due to users' demand for content during their free time on the road. This increased traffic leads to delays for users as it burdens the limited wireless communication capacity of base stations (BSs). While roadside units (RSUs) can alleviate this burden, their deployment costs contribute to the existence of outage zones. Existing precaching schemes that utilize one-hop vehicles are unable to minimize these outage zones due to an insufficient number of candidate vehicles. Therefore, we propose a Mobility-based Multi-hop Content Precaching Scheme for Connected and Cooperative Vehicular Networks (CCVNs). We limit the number of hops and the selection of precaching vehicles to minimize overheads and prevent excessive delays from frequent handovers. By selecting single-hop precaching vehicles, connectionless time is reduced because the requester vehicle can receive the requested content within the outage zone. Additionally, by selecting multi-hop precaching vehicles, we minimize the dark area time experienced by the requester vehicle. Our simulation results demonstrate that the proposed scheme effectively reduces both connectionless time and dark area time, thereby minimizing the outage zone compared to two representative existing schemes, Adaptive Carry-Store-Forward (ACSF) and Set Ranking-based Precaching (SRP), both based on single-hop communications.

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